LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	20
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Safety precautions to be observed while working on advanced electronic systems.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Safety precautions to be observed while working on advanced electronic systems.
5. Specific Objectives	20.7.1.3 Trainee will be able to observe safety precautions.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of ESD wrist strap usage, inspection of tool grounding, and verification of power isolation procedures.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary risk when handling sensitive electronic components?
(b) Why must power be isolated before working on avionics?
(c) What personal protective equipment is required for ESD safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Electrostatic Discharge (ESD) hazards • Grounding and bonding procedures • Power isolation protocols • Tool and equipment safety checks	L & Ds	PPT, IB	15 min
(b)	• Handling of high-voltage circuits • Use of insulated tools • Lockout/Tagout procedures • Emergency shutdown protocols	L & Ds	PPT, PDF	20 min
(c)	• Work area cleanliness standards • Cable management and routing • Personal hygiene and contamination control • Reporting of anomalies	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the correct sequence for donning an ESD wrist strap.
(b) Describe the steps required to safely isolate power on an avionics unit.
(c) List three critical safety checks before reconnecting a high-voltage system.

12. Home Assignment and Follow Up	Prepare a brief report on the specific safety precautions required for the next lesson plan (LP 02) regarding system testing procedures.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min